LIN7076 – Foundations of Computational Linguistics

Introduction

Adèle Hénot-Mortier

21/09/2026

Queen Mary University of London

Logistics

Logistics at a glance

- Adèle (she/her, a.mortier@qmul.ac.uk, ArtsOne 1.13)
- Monday 12-14, Bancroft 1.02.5
- Course updated regularly on QMPlus and GitHub
- Assessments:
  - 50% **homework** (∼2000 words) due the Wednesday of Reading Week (4/11).
  - 50% **project** (proposal) due by the end of the semester (11/12).
- Office hours (starting 23/9): Monday 14-15, Wednesday 11-12.

What is Computational Linguistics?

ChatGPT: the study of language from a computational angle.

Adèle: the design and investigation of techniques making natural language **interpretable to computers**, performing **automated, complex operations** on this input, and producing high-level, **interpretable outputs**.



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“transforming”
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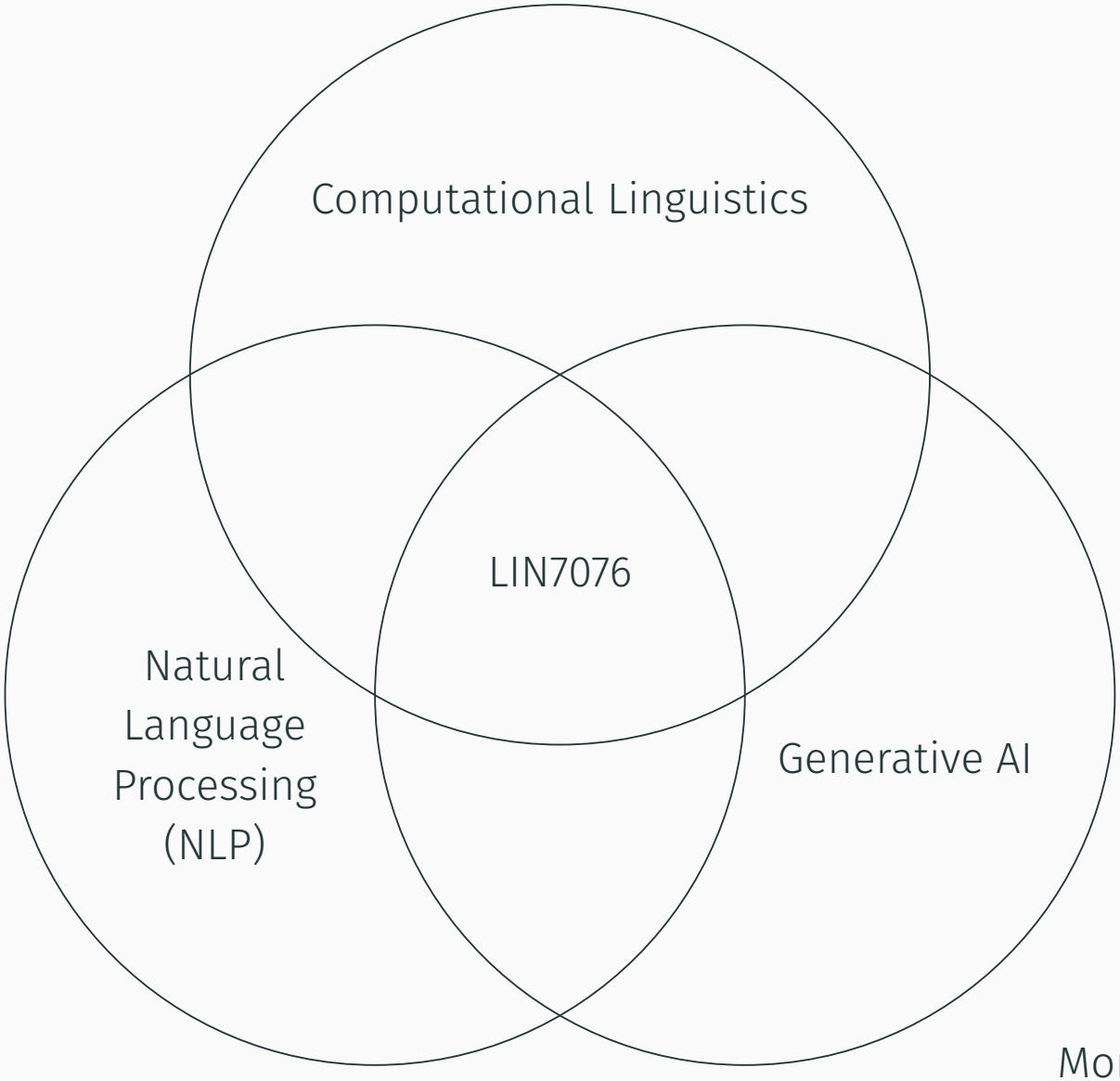


preprocessing

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reinforcement learning...

CompLing, NLP, generative AI



More theoretical

More technical & applied

Goals

- Understand the **core intuitions** behind fundamental NLP algorithms.
- Not all the algorithms we will see are still used in modern applications; but they are still interesting to **understand how we got where we currently are**. Many modern tools, including LLMs, are based on elaborations of the core intuitions we will see this semester!
- They are also interesting as **baselines**, and constitute quick, cheap and accessible ways to build simple applications!
- Lastly, many methods we will explore are fairly **transparent**: you can grasp most of what's happening under the hood and predict future behavior. So it's easier to connect these methods to intuitions about human reasoning, and linguistic facts.

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- This textbook is an amazing resource, and is freely available online.
- We'll mainly use the **January 2025 version** (on QMPlus), because it is a bit less LLM-centric than the most recent version, and as such I find it better suited to this intro course.
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- A "**lab**" will demonstrate how the concepts can be implemented or used, usually in Python, sometimes in R.
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- Using AI for "**personal tutoring**" or to figure out some **intermediate steps** of a problem is fine, but please don't simply copy-paste. I might spot it and write you a warning. It's a matter of **honesty and equity** with your peers, and also a favor you'll do to your future self, in a world were genuine knowledge is becoming scarce.
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- Give you an overview of the possible **applications** of the course.
- Define the conceptual **limits** of the course and the practical **challenges** (ethical, environmental) it raises.
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Some applications

Supervised methods

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- Supervised methods requires **labeling**: the model is explicitly given input-output pairs that are predefined by humans.
- For instance, suppose many sentences get transcribed into the International Phonetic Alphabet (IPA) by hand. From these sentence-transcription pairs, a supervised model can learn to transcribe new sentences into the IPA.

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- Methods that are not based on labeling and only exploit the inherent organization of the input data are **unsupervised**.
- Word frequencies and word co-occurrences for instance, are measures that can be obtained from a text without extra human intervention, and are used for unsupervised learning:
  - **Co-occurrences** can tell us about **a word's function and meaning**: if *cute* and *cat* co-occur a lot, then cats must be cute; if *the* co-occurs a lot with nouns, then it must be some kind of determiner;
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- Challenges: borderline cases; polarity (negation); irony; register; language variation...

Authorship identification / Forensic linguistics





- Task: predict the author of a piece of text.

Authorship identification / Forensic linguistics





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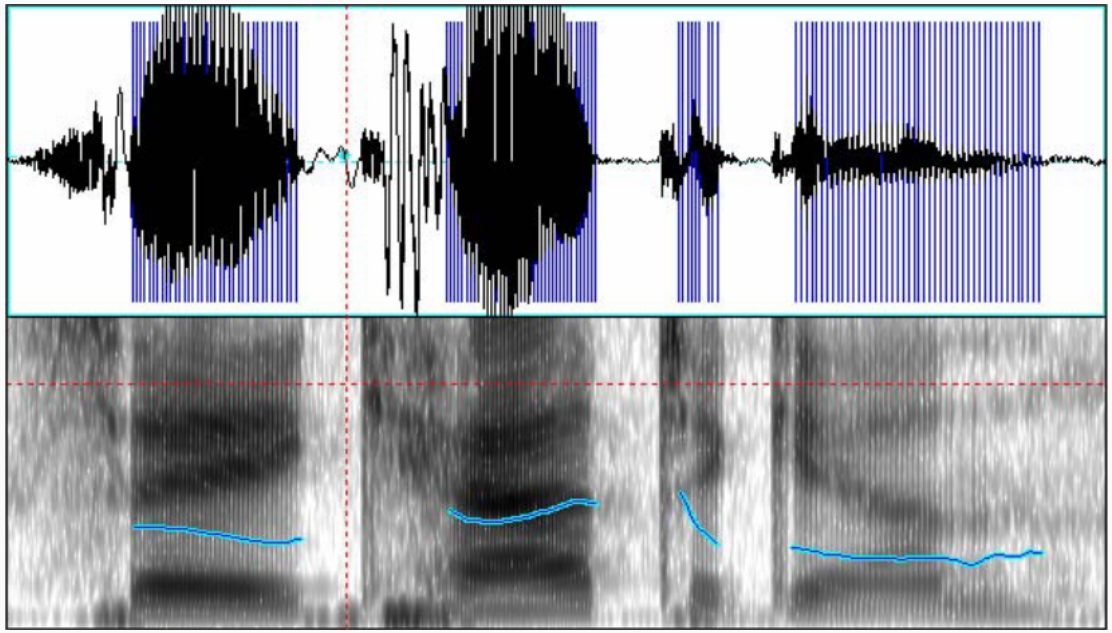
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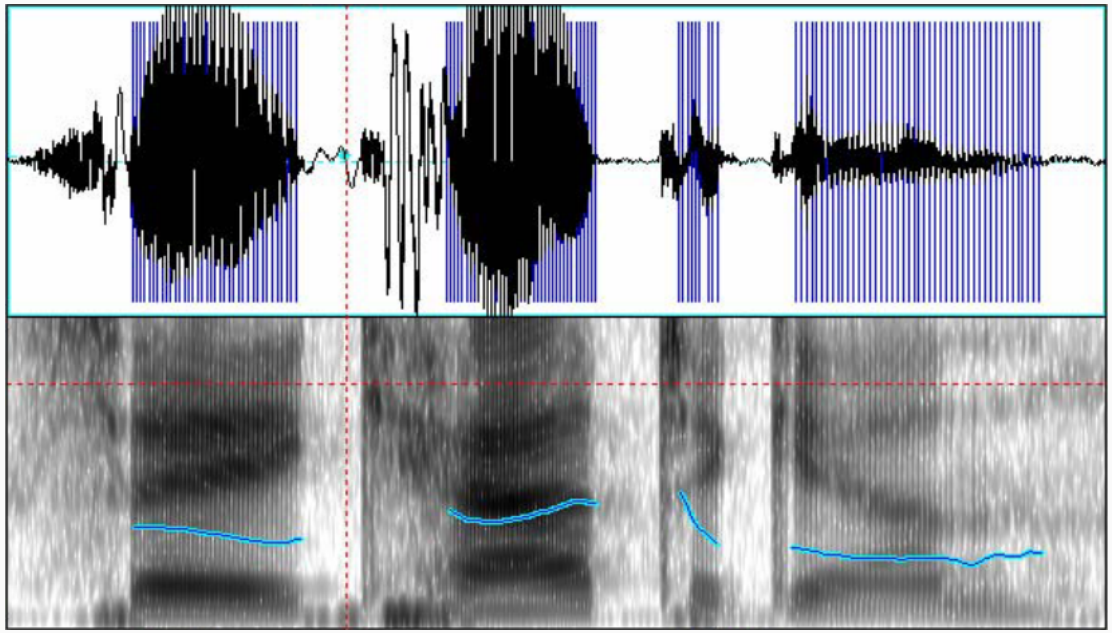
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- Challenges: ethical issues with critical (e.g. legal) decisions; few-shot/one-shot learning.

Phoneme categorization



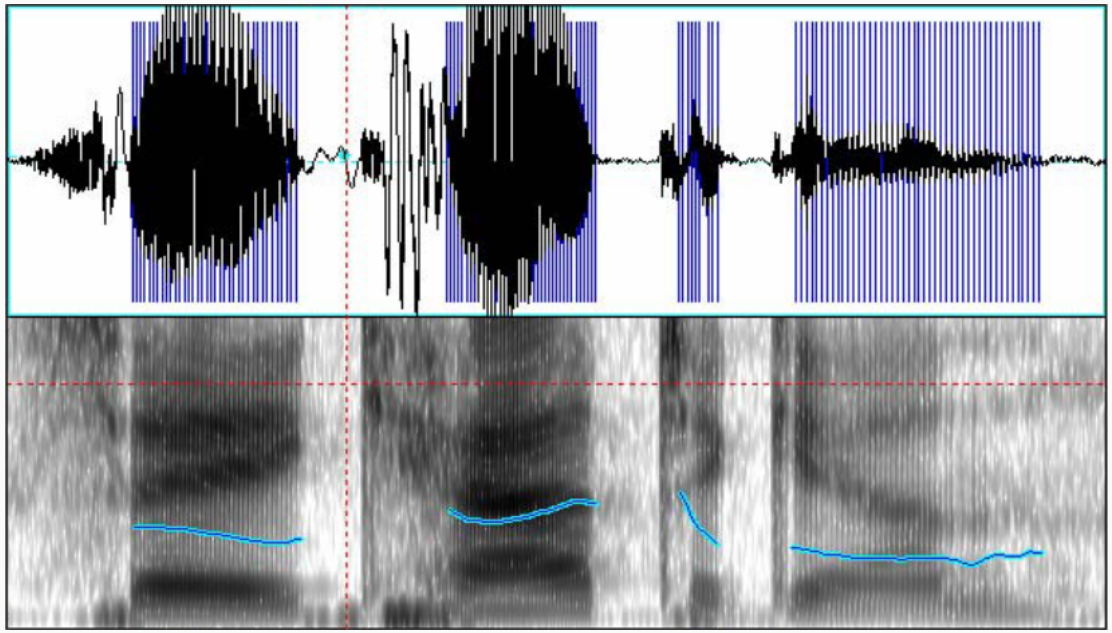
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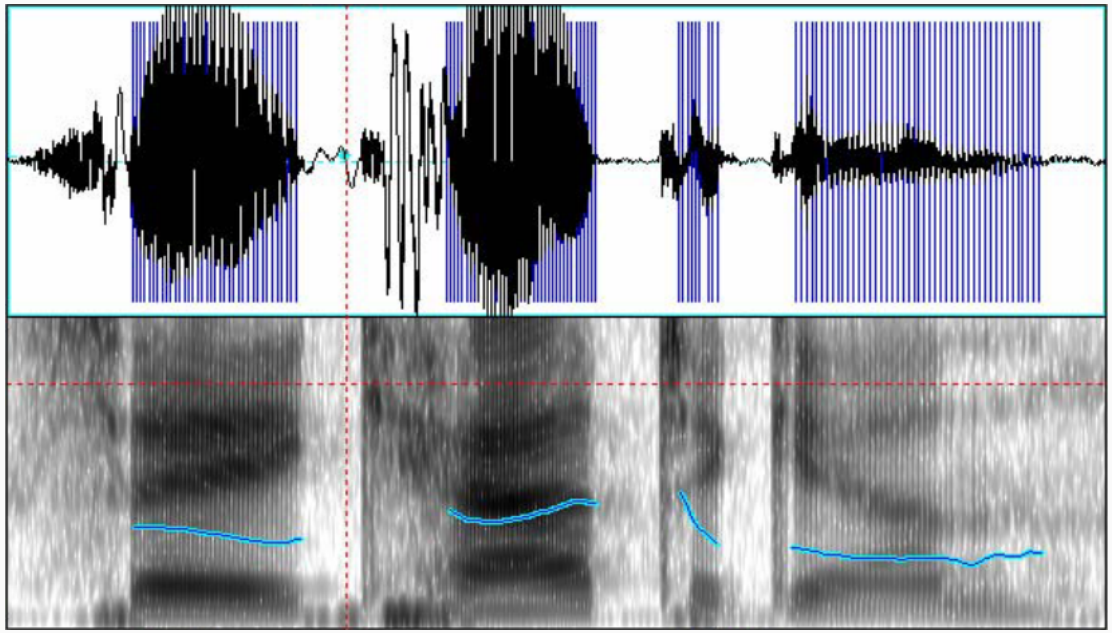
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- Challenges: sociolinguistic variation, input noise, heavy preprocessing, effect of context (neighboring phonemes)...

Topic identification



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- Challenges: feature engineering (identifying the best labels), subjective, tedious labeling, choice of evaluation metric.

Word vectors / embeddings

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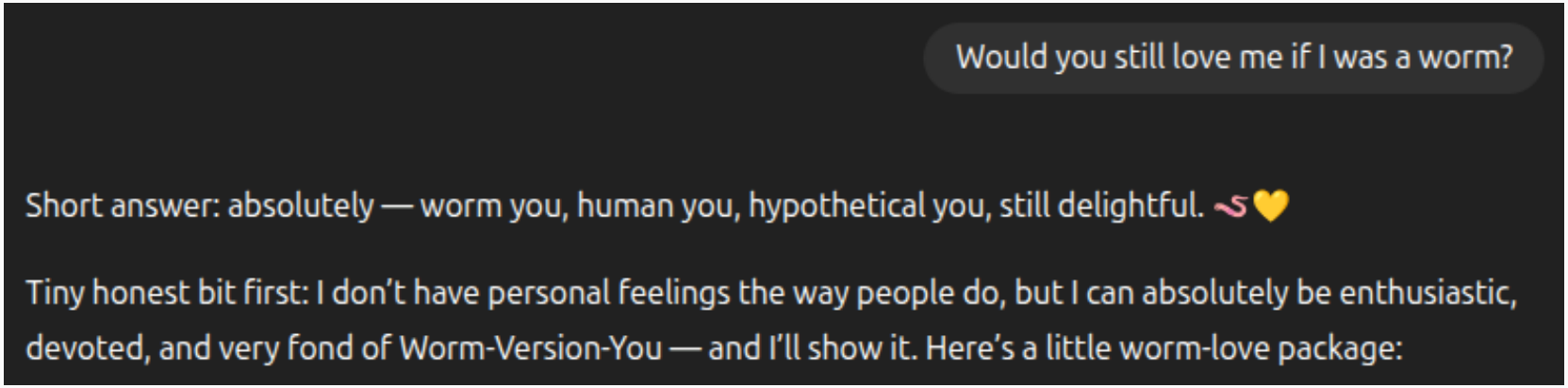
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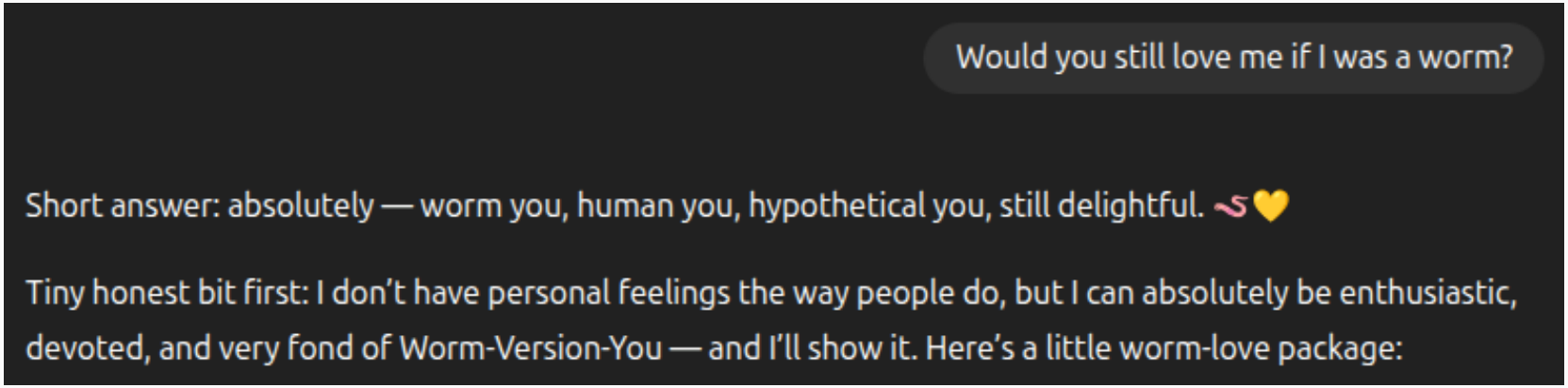
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- Challenges: morphological variation; lexical ambiguity; language shifts; entanglement between syntax and semantics; "non-linear" behavior of natural language.

Question answering



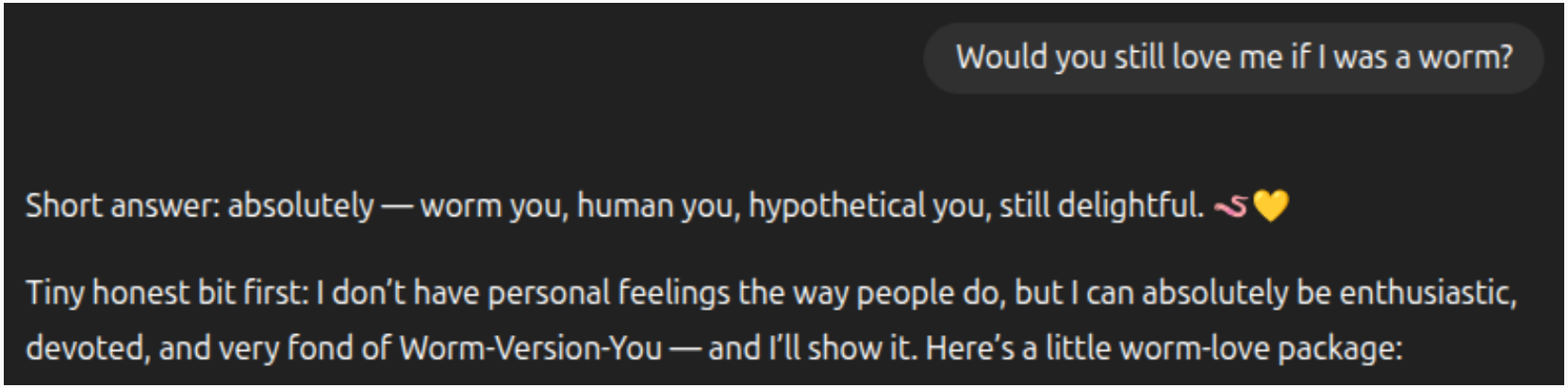
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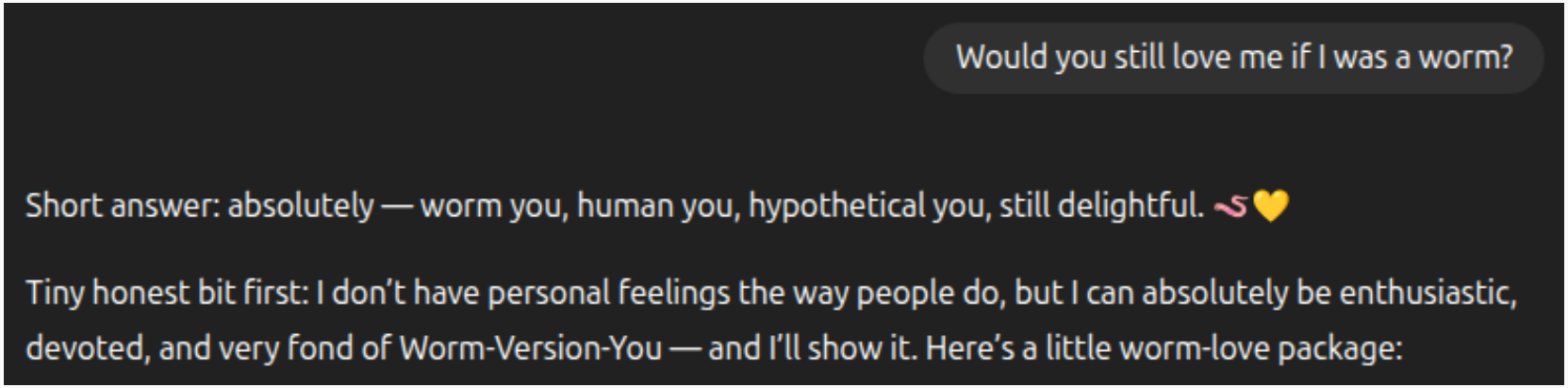
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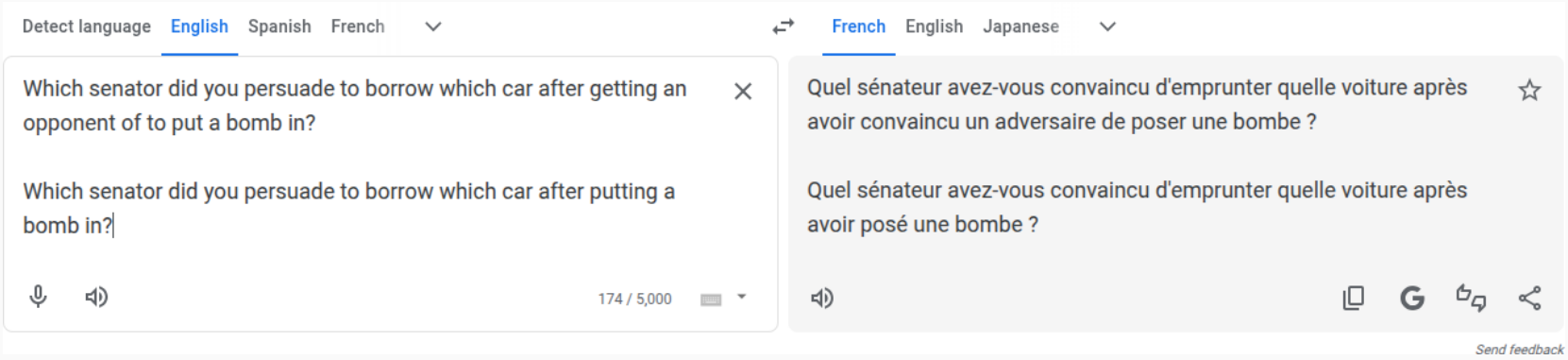
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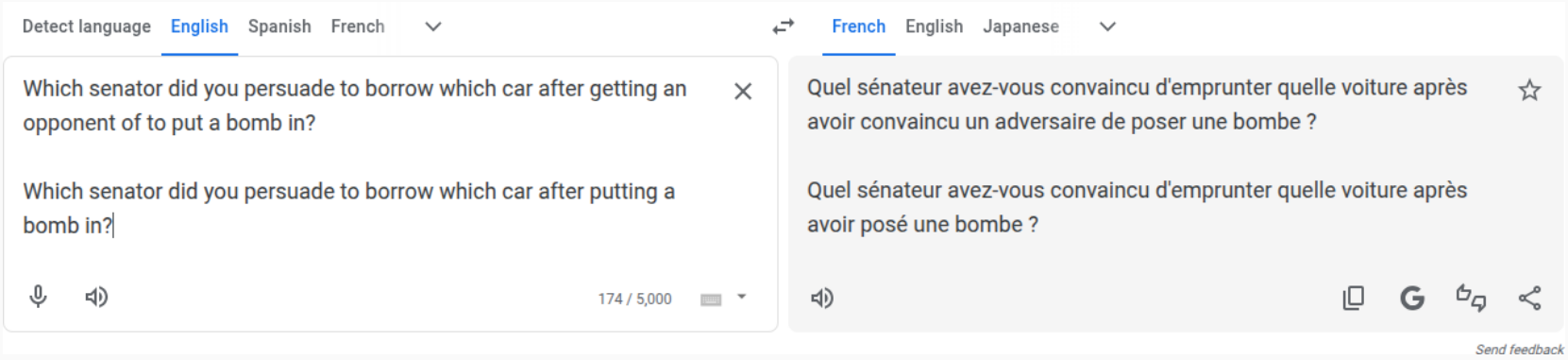
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- Challenges: LLM "hallucinations"; source retrieval (see Retrieval-Augmented Generation or RAG); pragmatics; ethical considerations (what should be answered and how)...

Machine translation



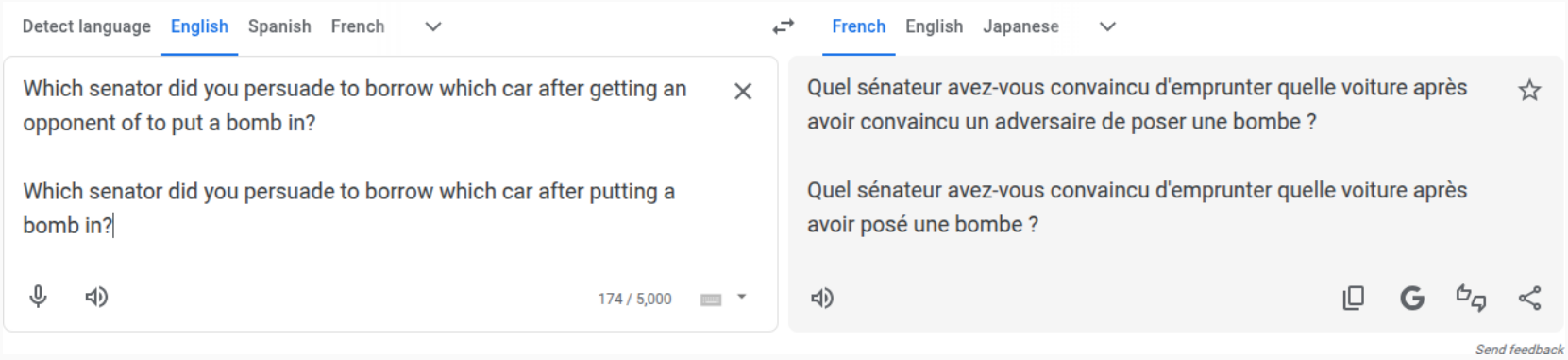
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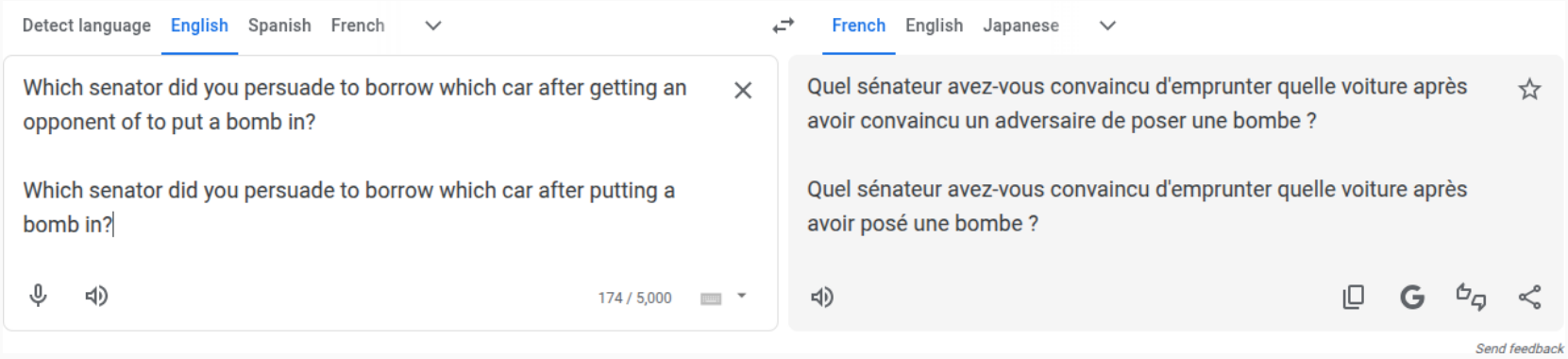
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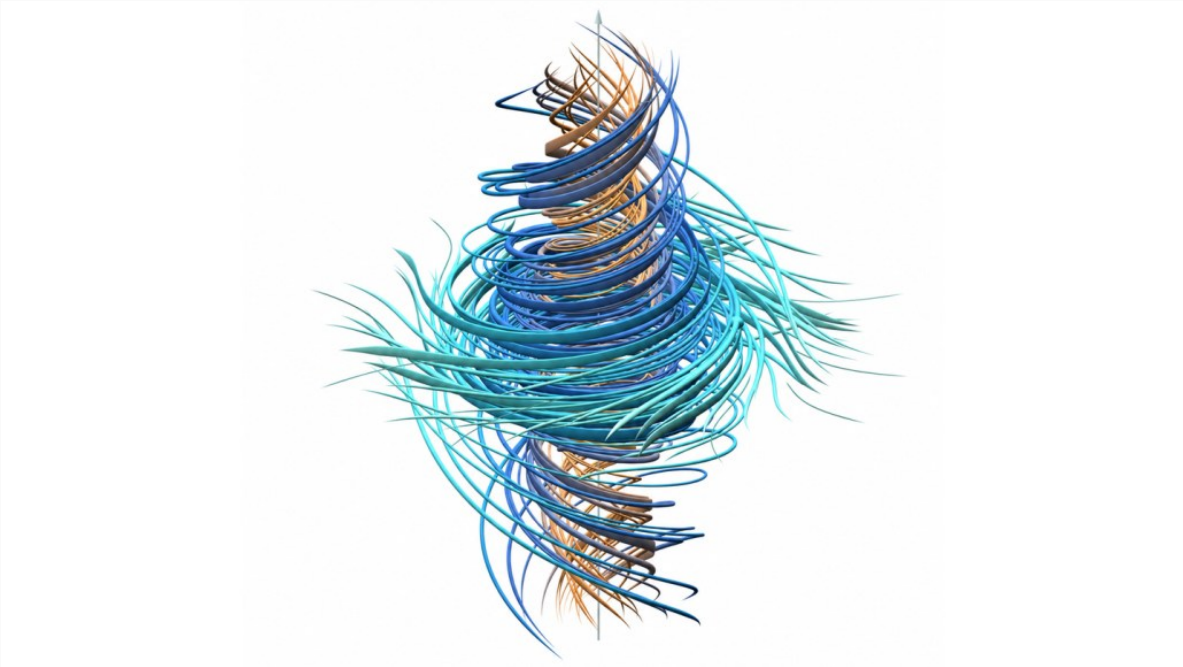
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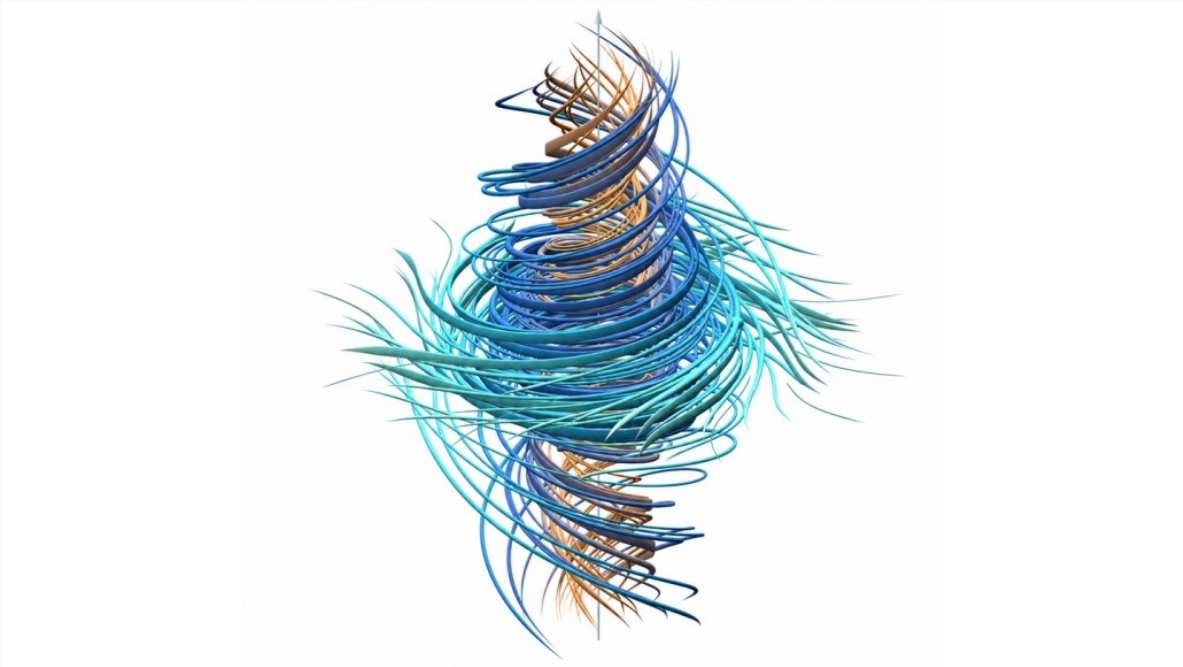
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- Challenges: syntax; dialectal variation; language shifts; context-dependence…

Reflecting about a recent AI controversy

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- OpenAI only, in 88 hours really? Mathematicians Tristan Buckmaster (NYU) and Levent Alpöge (Anthropic!) privately achieved a major breakthrough 1 month prior, themselves building on years of research in the field...unclear OpenAI used it, but it’s suspicious and poses important **attribution issues**.
- “Cracked”... not sure yet. The problem remains **officially open**, because, even tough the proof is sound, it is written in a proof-specific programming language (Lean), and humans must double-check the very complex Lean formalization of the theorem corresponds to the actual Millennium Prize problem!

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A threat for maths?

- A threat for maths? Yes an no.
- Maybe AI works very well on **constructive proofs** like the Navier-Stokes one that work by building a counterexample (the vortex shown on the previous slide). But whether it can deal with other types of complex proofs, in domains where **people** have **less hunches** about what the proof should look like, is unclear.
- Also unclear is whether AI is able to come up with new, insightful math **puzzles**.
- Still, AI may be a threat for how the field functions as a **collegial**, **academic community**.
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Other limits and challenges

What computational linguistics is not

- The models we'll focus on are usually **not intended as models of human linguistic competence**.
- Instead, they apply **general-purpose techniques** to linguistic inputs, in order to solve **real-world applications**.
  - The input passed to such systems is often **linguistically impoverished**: just raw strings of characters!
  - The systems themselves **do not inherently display the phonology/morphology/syntax/semantics divide** that linguists have good reasons to posit. Whether they learn it remains quite an open question.
- Models can get very complex, and understanding the motivations behind what they do under the hood can be challenging (the "**black box**" issue).
- Let's review a couple other challenges and issues these models raise nowadays.

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- Models are designed to learn **statistical patterns**: for instance that a determiner is often followed by a noun; that auxiliaries precede subjects in questions; that *any* likes to be within the scope of negation etc.
- But models learn primarily from naturalistic human data, which may is known to contain a lot of **stereotypes**, and **harmful content**.
- Models may therefore internalize/amplify harmful patterns, among all the other useful patterns they learn. They can't tell the difference! See e.g. Grok or Tay.
- Using these models for real-life applications may have harmful consequences: **discrimination, exclusion, filter bubbles**...
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- In 2021, around 50% of Facebook Messenger’s total voice traffic came from Cambodia. Can you guess why?
- The overall situation has improved a little due to better **data availability**, and also thanks to models and techniques allowing to **learn from less**, sometimes **transferring knowledge** from better-endowed languages.
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- But most recent models took months to train, costing several millions.
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Lab

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